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Editors

Mathematical Programming Computation

Dear Editors,

Please consider the manuscript “*Salem–Spencer sets as a cross-solver benchmark: decomposition, certification, and split-policy effects at $r_3(212)$* ” for publication in *Mathematical Programming Computation*.

The manuscript presents a reproducible computational study and benchmark family built from the open frontier decision problem for OEIS A003002. It does not claim to resolve $r_3(212)$. Instead, it develops and evaluates a witness-informed cube decomposition, quantifies the effect of problem-specific window-cardinality cuts, compares CP-SAT, LP/MIP, and CDCL formulations, and releases a graded instance library with machine-checkable proof artifacts.

The study contains five results aligned with MPC’s emphasis on practical computation, comparative testing, reusable software, and verification:

- Six monolithic 24-hour baselines fail. A matched five-policy ablation exposes a cover-size versus conquer-cost tradeoff: global AP-degree reduces the exact emitted cover from 12,582,912 to 96,847 cubes, but leaves individually harder subproblems under the historical CP-SAT cap. A standalone C implementation independently reproduces the complete cover.
- A two-stage CDCL survey closes 6,045 of 6,071 broad residuals (99.57%) with no feasible result, while 26 released instances remain as a sharply defined challenge tier.
- On 20 byte-identical hard-pocket formulas run sequentially on matched AMD EPYC 7763 nodes, CaDiCaL 3.0.0 and kissat 4.0.4 both close every instance. Kissat wins all 20 pairs, with geometric-mean CaDiCaL-to-kissat time ratio 1.477 (bootstrap 95% interval 1.332–1.655).
- Six solve-time-stratified survivors from the alternative global-degree cover reproduce the survey CNF hashes, verify through DRAT-to-LRAT conversion, and are accepted by the formally verified `cake_lpr` checker. Separate end-to-end regressions recover the known exact values $r_3(80) = 22$, $r_3(90) = 24$, and $r_3(100) = 27$ from independently verified lower witnesses and `cake_lpr`-checked upper certificates.
- Optimization-form HiGHS experiments show that window inequalities reduce the T1b node count by approximately 81% and tighten every tested LP and MIP upper bound to the decision threshold 44, but not below it.

Source code and benchmark generators are available at https://github.com/memoatwit/erdos_1194/tree/mpc-submission-v1.4/erdos_r3. Versioned artifacts are archived at <https://doi.org/10.5281/zenodo.21413746>. The journal-submission version of the archive includes formulas, solver outputs, proof objects, hashes, an independent model audit, and machine-readable provenance, including the six global-degree certificate bundles. The preprint is available as arXiv:2606.04016. The manuscript is not under consideration at another journal.

For Technical Editor review, `ARTIFACT_REPRODUCIBILITY.md` maps each claim to its data and reproduction command. The $N = 80$ exact-value regression is the fastest end-to-end route and runs in minutes on a workstation; the archived certificates can also be checked with `drat-trim` and `cake_lpr` without re-solving the instances.

The paper retains and explicitly refutes a working hypothesis formed during the campaign. We believe this is methodologically useful: it demonstrates why unresolved instances, solver configurations, and proof objects should be archived and retested under controlled pipelines.

Generative-AI use is disclosed in the manuscript. The author made all final scientific and experimental decisions, independently checked the numerical and mathematical claims, and takes full responsibility for the work.

Sincerely,

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